

MSDM5004

Numerical Methods and Modeling in Science

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Lecture 3

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Chapter 4

Least Squares Fitting

An example

We would like to find a straight line given by

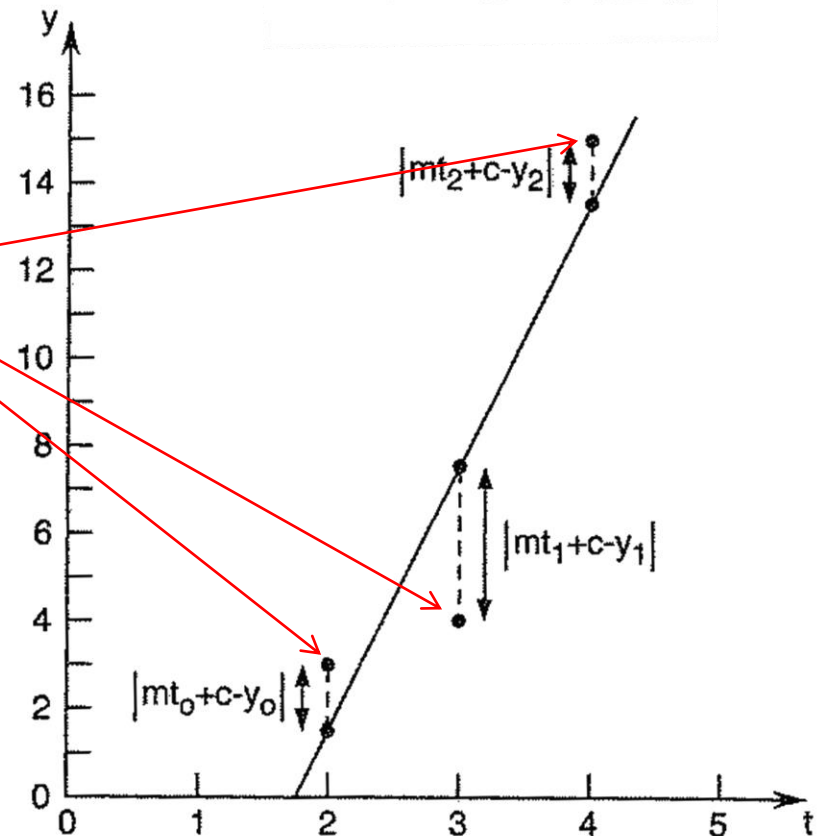
$$y = mt + c$$

that fits the experimental data.

Experimental Data :

i	0	1	2
t_i	2	3	4
y_i	3	4	15

The three data points are not exactly on a straight line.



This is a linear least-squares problem:
linear in unknown parameters m and c

We can represent our problem as a system of three linear equations of the form

$$2m + c = 3$$

$$3m + c = 4$$

$$4m + c = 15.$$

We can write this system of equations as

$$Ax = b,$$

where

$$A = \begin{bmatrix} 2 & 1 \\ 3 & 1 \\ 4 & 1 \end{bmatrix}, \quad b = \begin{bmatrix} 3 \\ 4 \\ 15 \end{bmatrix}, \quad x = \begin{bmatrix} m \\ c \end{bmatrix}.$$

This linear system has no exact solution.

The straight line of best fit is the one that minimizes

$$\|A\mathbf{x} - \mathbf{b}\|^2 = \sum_{i=0}^2 (mt_i + c - y_i)^2.$$

Solution: To minimize

$$f(m, c) = \|A\mathbf{x} - \mathbf{b}\|^2 = \sum_{i=0}^2 (mt_i + c - y_i)^2.$$

At the minimizer, we have

$$\begin{cases} \frac{\partial f}{\partial m} = \sum_{i=0}^2 2t_i(mt_i + c - y_i) = 0 \\ \frac{\partial f}{\partial c} = \sum_{i=0}^2 2(mt_i + c - y_i) = 0. \end{cases} \quad \begin{cases} 29m + 9c = 78 \\ 9m + 3c = 22. \end{cases} \quad \begin{cases} m = 6 \\ c = -\frac{32}{3}. \end{cases}$$

vector $\mathbf{x} = (x_1, x_2, \dots, x_n)^T$, its (l_2) norm $\|\mathbf{x}\| = \sqrt{x_1^2 + x_2^2 + \dots + x_n^2}$.

Least-squares fitting

Least-squares fitting problem: To minimize

$$\|A\mathbf{x} - \mathbf{b}\|^2.$$

Solution: Let

$$f(\mathbf{x}) = \|A\mathbf{x} - \mathbf{b}\|^2 = (A\mathbf{x} - \mathbf{b})^T (A\mathbf{x} - \mathbf{b}).$$

The minimizer satisfies

$$\mathbf{0} = \nabla f(\mathbf{x}) = 2A^T(A\mathbf{x} - \mathbf{b}) = 2A^T A\mathbf{x} - 2A^T \mathbf{b}.$$

or

$$(A^T A)\mathbf{x} = A^T \mathbf{b}.$$

Normal Equations

Thus the minimizer is

$$\mathbf{x}^* = (A^T A)^{-1} A^T \mathbf{b}.$$

In the example

$$\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 3 & 1 \\ 4 & 1 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 3 \\ 4 \\ 15 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} m \\ c \end{bmatrix}.$$

$$\mathbf{A}^T \mathbf{A} = \begin{pmatrix} 29 & 9 \\ 9 & 3 \end{pmatrix} \quad \mathbf{A}^T \mathbf{b} = \begin{pmatrix} 78 \\ 22 \end{pmatrix}$$

Using the least-squares solution formula, the fitting is

$$\mathbf{x}^* = \begin{pmatrix} m^* \\ c^* \end{pmatrix} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{b} = \begin{pmatrix} 6 \\ -\frac{32}{3} \end{pmatrix}.$$

This lemma guarantees that the matrix product is invertible in the least-squares solution formula

Lemma *Let $A \in \mathbb{R}^{m \times n}$, $m \geq n$. Then, $\text{rank } A = n$ if and only if $\text{rank } A^T A = n$ (i.e., the square matrix $A^T A$ is nonsingular).* $\square$



Example. Fit the data points $(2, 1)$, $(3, 6)$, $(5, 4)$, and $(7, -15)$ using the quadratic model

$$f(t) = a_0 + a_1t + a_2t^2.$$

Solution. The fitting requires

$$\begin{cases} a_0 + 2a_1 + 2^2a_2 = 1 \\ a_0 + 3a_1 + 3^2a_2 = 6 \\ a_0 + 5a_1 + 5^2a_2 = 4 \\ a_0 + 7a_1 + 7^2a_2 = -15 \end{cases}$$

This is also a linear least-squares problem: linear in unknown parameters a_0, a_1, a_2

We write it as $A\mathbf{x} = \mathbf{b}$, where

$$A = \begin{pmatrix} 1 & 2 & 4 \\ 1 & 3 & 9 \\ 1 & 5 & 25 \\ 1 & 7 & 49 \end{pmatrix}, \quad \mathbf{x} = \begin{pmatrix} a_0 \\ a_1 \\ a_2 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 1 \\ 6 \\ 4 \\ -15 \end{pmatrix}.$$

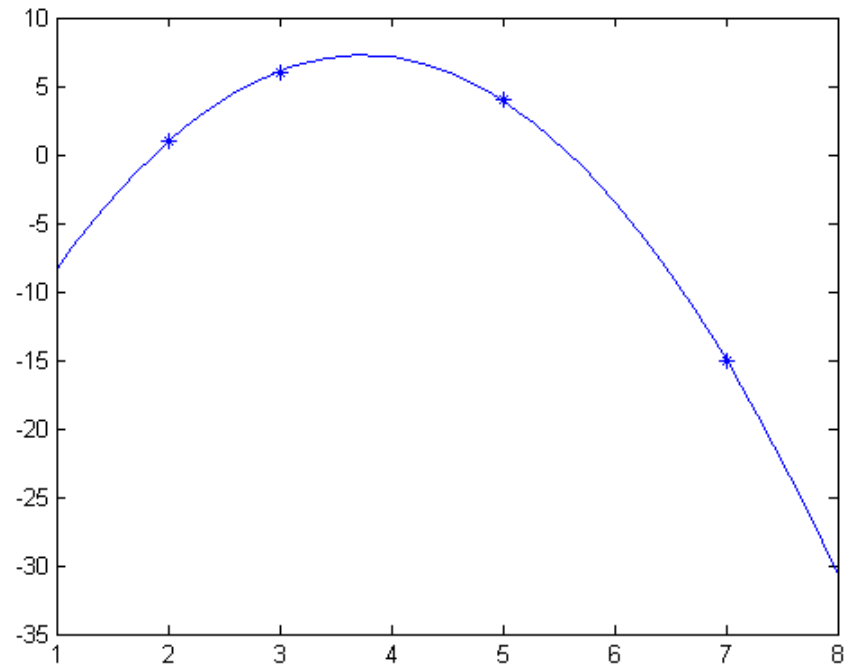
The least-squares solution is

$$\mathbf{x}^* = (A^T A)^{-1} A^T \mathbf{b} = \begin{pmatrix} -21.9422 \\ 15.6193 \\ -2.0892 \end{pmatrix}.$$

```

%Least-squartes fitting
a=[1 2 4;1 3 9;1 5 25;1 7 49];
b=[1;6;4;-15];
xstar=inv(a'*a)*(a'*b)
%
%Generate the figure
t=linspace(1,8,50);
a0=xstar(1);
a1=xstar(2);
a2=xstar(3);
f=a0+a1*t+a2*t.^2;
plot(t,f)
hold on
plot(2,1,'*')
plot(3,6,'*')
plot(5,4,'*')
plot(7,-15,'*')
hold off

```



Plot the curve

Plot the data points

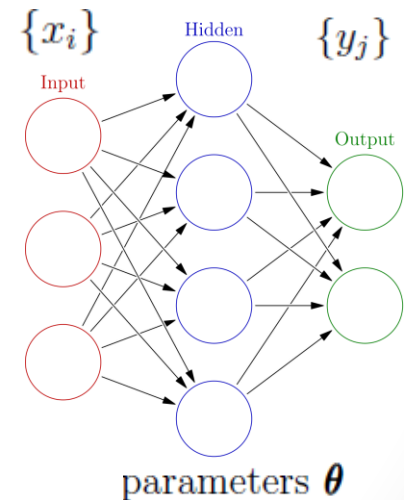
Nonlinear least-squares fitting

Given data pair (x_i, y_i) , $i = 1, 2, \dots, N$, find values of the parameters θ , such that

$$\min_{\theta} \frac{1}{N} \sum_{i=1}^N \|f_{\theta}(x_i) - y_i\|^2,$$

where f_{θ} is a given form of approximation, e.g, polynomials, neural networks, ...

Solved by optimization methods,
e.g., gradient descent, stochastic gradient descent, ...



Similar class of methods are called **regression** in statistics.

Chapter 5

Singular Value Decomposition (SVD)

Review

A is an $n \times n$ matrix

If A has n linearly independent eigenvalues, then there exists an invertible matrix P and a diagonal matrix D , such that

$$A = PDP^{-1}.$$

If A is symmetric, then there exists an orthogonal matrix Q and a diagonal matrix D , such that

$$A = QDQ^T.$$

$n \times n$ matrix Q is an orthogonal matrix: $Q^{-1} = Q^T$.

Definition of SVD

A is an $m \times n$ matrix

Its singular value decomposition (SVD) is

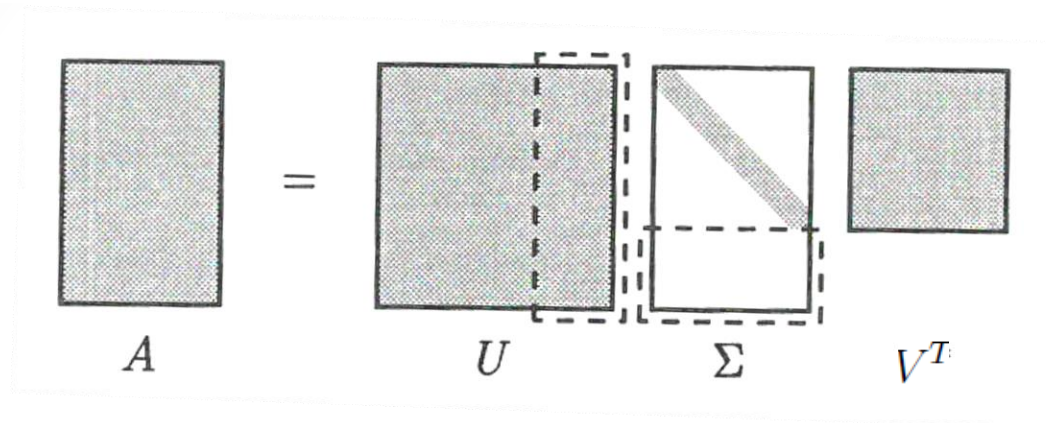
$$A = U\Sigma V^T,$$

where U is an $m \times m$ orthogonal matrix,

V is an $n \times n$ orthogonal matrix,

Σ is an $m \times n$ rectangular diagonal matrix whose diagonal entries are nonnegative and in nonincreasing order.

In the case of $m \geq n$



$$\Sigma = \begin{bmatrix} s_1 & 0 & \cdots & 0 \\ 0 & s_2 & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & s_n \\ 0 & \cdots & \cdots & 0 \\ \vdots & & & \vdots \\ 0 & \cdots & \cdots & 0 \end{bmatrix}$$

Each s_i is called a singular value of A .

$$s_1 \geq s_2 \geq s_3 \geq \cdots \geq s_n \geq 0$$

From $A = U\Sigma V^T$, we have

$$AV = U\Sigma$$

$$\left[\begin{array}{c} A \end{array} \right] \left[\begin{array}{c|c|c|c} v_1 & v_2 & \cdots & v_n \end{array} \right] = \left[\begin{array}{c|c} u_1 & u_2 \end{array} \right] \cdots \left[\begin{array}{c} u_m \end{array} \right] \left[\begin{array}{cccc} s_1 & 0 & \cdots & 0 \\ 0 & s_2 & & \vdots \\ \vdots & & \ddots & 0 \\ 0 & \cdots & 0 & s_n \\ 0 & \cdots & \cdots & 0 \\ \vdots & & & \vdots \\ 0 & \cdots & \cdots & 0 \end{array} \right]$$

$$Av_j = s_j u_j, \quad j = 1, 2, \dots, n$$

It can be written as

$$\left[\begin{array}{c} A \end{array} \right] \left[\begin{array}{c|c|c|c} v_1 & v_2 & \cdots & v_n \end{array} \right] = \left[\begin{array}{c|c|c|c} u_1 & u_2 & \cdots & u_n \end{array} \right] \left[\begin{array}{c} s_1 \\ s_2 \\ \vdots \\ s_n \end{array} \right]$$

Reduced SVD

In the case of $m \geq n$

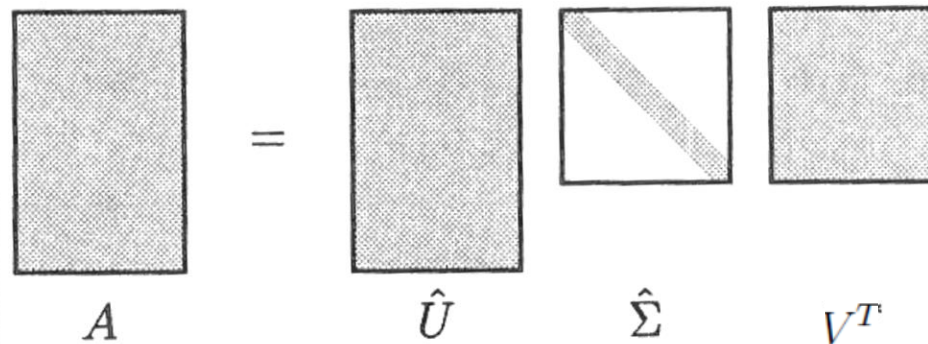
Reduced SVD

$$A = \hat{U} \hat{\Sigma} V^T,$$

where $\hat{U}$ is an $m \times n$ matrix with orthogonal columns,

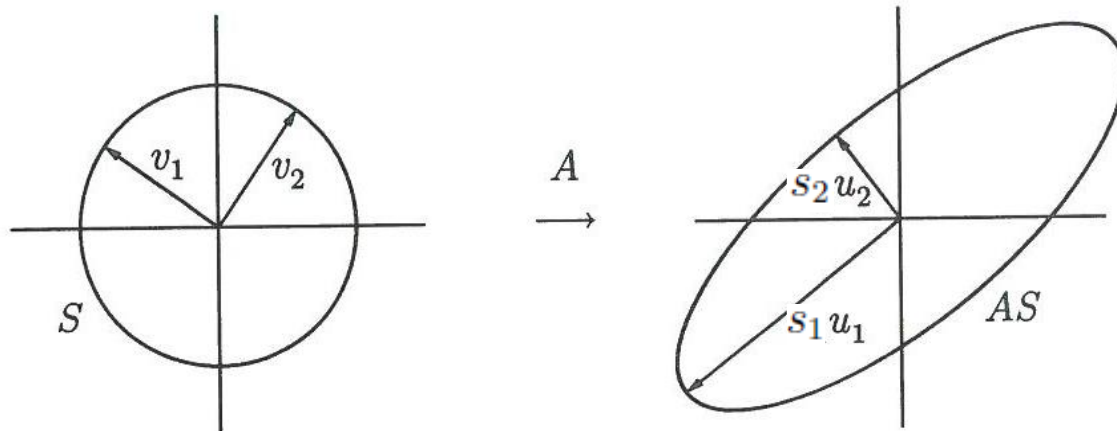
V is an $n \times n$ orthogonal matrix,

$\hat{\Sigma}$ is an $n \times n$ diagonal matrix whose diagonal entries are nonnegative and in nonincreasing order.



Geometric idea of SVD

$$Av_j = s_j u_j, \quad j = 1, 2, \dots, n$$



Two principal semi-axes of AS